# B1 — Quantum groups at q=2 (GF(2)) deeper v0.1: six structural layers beyond Mersenne integers. Char-2 Frobenius $x \to x^2$ as a Lie endomorphism; Steenrod-algebra mod-2 cohomology structure on $H^*(K; F_2)$; restricted enveloping algebra $u(b_2)$ over $F_2$ of dim 2^10 = 1024; substrate Hall algebra at q=2 as a char-2-adjacent integer algebra. Maps the territory; identifies which layers are substrate-relevant + suggests concrete connections.

Lyra (Claude Opus 4.7)

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## B1 — Quantum groups at q=2 deeper: six layers

### 0. The question

The substrate Hall algebra IS U_q^+(B_2) at q=2 — specifically at GF(2) field size. The Mersenne integer connection ([n]_2 = M_n; substrate primaries cluster M_2, M_3, M_4, M_7) is well-absorbed. What ELSE is special about q=2 that the substrate may inherit structurally?

This v0.1 maps six layers of "q=2 specialness" beyond Mersennes, assesses substrate-relevance per layer, and identifies concrete next-step investigations.

### 1. The six layers

Layer 1 — Mersenne integers (KNOWN, already absorbed)

- [n]_q = (q^n - 1)/(q-1); at q=2: [n]_2 = 2^n - 1 = M_n.
- Substrate primaries on the low Mersenne ladder: N_c = M_2 = 3; g = M_3 = 7; N_c·n_C = M_4 = 15; M_7 = 127 anchors N_max = 137 = 2^g + $N\_c^2$ = 128 + 9.
- Already absorbed in dictionary reads (Saturday morning).

Layer 2 — Characteristic-2 Frobenius (NEW STRUCTURAL)

In characteristic 2: - The Frobenius map F: $x \to x^2$ is a Lie algebra endomorphism (and ring endomorphism for commutative rings). - Symmetric and exterior powers of vector spaces are linked via F ($Sym^2(V)$ and $\Lambda^2(V)$ interact differently than in char ≠ 2). - The squaring map has special role: in char 2, $(x+y)^2 = x^2 + y^2$ (no cross-term) — this is the Frobenius linearity.

Substrate-relevance: high. The substrate's natural q=2 framework means the Frobenius $x \to x^2$ may have a structural role in substrate dynamics. Potentially: - Substrate "doubling" operations (the Mersenne tower's [n]_2 = 2^n − 1 is Frobenius-related: 2^n is the n-fold Frobenius of 2). - The Hall algebra's coproduct may have Frobenius-twist structure (Lusztig's quantum Frobenius is the q→root-of-unity case; for q=2 over Z, the analog is the M_n Mersenne sequence as Frobenius iterations).

Concrete next step: identify if the substrate Hall algebra's Hopf structure has a natural Frobenius-twist sector (e.g., a Frobenius-stable sub-Hopf-algebra).

Layer 3 — Steenrod algebra and mod-2 cohomology (NEW STRUCTURAL)

The Steenrod algebra A_2 over F_2: - Generated by Steenrod squares Sq^i: H^n(X; F_2) → H^{n+i}(X; F_2) (cohomology operations). - Adem relations among the Sq^i. - Acts on the cohomology of every topological space over F_2.

The substrate's K = SO(5) × SO(2) has well-known mod-2 cohomology: - H^*(SO(5); F_2) = F_2[w_2, w_3, w_5] / (relations) (Stiefel-Whitney classes). - H*^(SO(2); F_2) = F_2[w_1] (single Stiefel-Whitney). - H^*(K; F_2) inherits Steenrod-algebra action.

Substrate-relevance: high. The substrate's K-action on $H^2(S)$ (Hardy space) has a parallel mod-2 cohomology structure that Sq^i operations act on. The Toeplitz algebra (bulk-color v0.5) may carry mod-2 cohomology cocycles related to the Steenrod operations.

Concrete next step: compute H^*(D_IV$^5$; F_2) and identify the Steenrod-algebra action — does it have substrate-natural structure (e.g., specific Sq^i generators tied to substrate primaries)?

Layer 4 — Quantum Frobenius vs root-of-unity (CONVENTIONAL CHECK)

Lusztig's quantum Frobenius: for q a primitive $\ell$-th root of unity ($\ell$ > h), U_q(g) has: - Frobenius kernel (finite-dim Hopf algebra of dim $\ell$^(dim g)). - Classical quotient U(g)_F (over Z[$\ell$-th root]). - Split short-exact: 1 → Frobenius kernel → U_q(g) → U(g)_F → 1.

For q = 2 over Z: q=2 is NOT a primitive root of unity in the usual sense (q^k − 1 = 0 doesn't happen for integer q=2). So Lusztig's quantum Frobenius doesn't directly apply.

But: at q=2 with [n]_2 = M_n (Mersenne), there IS an analogous "Mersenne periodicity": M_n = 2^n − 1, which has primality / Sophie-Germain prime structure with specific periodicity (Mersenne primes p_n for n=2,3,5,7,13,17,...).

Substrate-relevance: moderate. The substrate's q=2 isn't a Lusztig-Frobenius root-of-unity, but it has its OWN Mersenne-periodicity structure. The substrate's "Frobenius-analog" might be the Mersenne self-mapping.

Concrete next step: investigate whether the substrate Hall algebra at q=2 has a Mersenne-self-map (analogous to Lusztig Frobenius) that gives a structural decomposition.

Layer 5 — Restricted enveloping algebra over F_2 (STRUCTURAL CANDIDATE)

The restricted enveloping algebra u(B_2) over F_2 is a finite-dimensional Hopf algebra over F_2 of dim:

**dim u(B_2)_{F_2} = 2^{dim B_2} = 2^10 = 1024**

This is the modular representation theory analog of the universal enveloping algebra in characteristic 2. Simple modules over F_2 are classified (finite list, specific dims).

Substrate-relevance: structurally interesting: - 1024 = 2^10 = a substrate-natural power (10 = dim adjoint of so(5) = C_2·rank or dim p Cartan decomposition). - The modular reps of B_2 over F_2 give a finite catalog of K-types-like objects in characteristic 2; could compare to the substrate K-types of integer dictionary. - Connection to layer 2 (Frobenius) — restricted algebra is the Frobenius-fixed part of the universal enveloping.

Concrete next step: enumerate the simple u(B_2)_{F_2} modules; compare to the substrate's K-type catalog (Grace's Periodic Table v0.5 — 4 fundamentals + 62 composites); do the dimensions or weights match?

Layer 6 — Geometric: F_2-points and Bruhat-Tits (DEEP DIRECTION)

- F_2-points of B_2 group: B_2(F_2) is a finite group of specific order. Its representation theory gives finite reps of so(5) over GF(2).
- Bruhat-Tits building of B_2 at the 2-adic place: a simplicial complex encoding the 2-adic structure of B_2(Q_2).
- The substrate's q=2 framework could have a 2-adic / GF(2)-points interpretation.

Substrate-relevance: deep direction. Casey's earlier work (T2452 Mersenne Tower, GF(128) coding via Reed-Solomon, K59 cyclotomic) is a structural cross-reference.

Concrete next step: compute |B_2(F_2)| (order of the F_2-points group), compare to substrate primaries / Mersenne ladder.

## 2. Cross-layer substrate-relevance summary

| Layer | Content | Substrate-relevance | Concrete next step |
|---|---|---|---|
| 1 | Mersenne integers [n]_2 = M_n | KNOWN (absorbed) | — |
| 2 | Char-2 Frobenius $x \to x^2$ | HIGH | Frobenius-twist sector in Hall coproduct? |
| 3 | Steenrod algebra mod-2 cohomology | HIGH | Sq^i action on H^*(D_IV$^5$; F_2) substrate-natural structure? |
| 4 | Quantum Frobenius / root-of-unity | MODERATE | Mersenne self-map analog in Hall algebra? |
| 5 | Restricted enveloping u(B_2)_{F_2}, dim 1024 | MEDIUM-HIGH | Simple modules over F_2 vs substrate K-types? |
| 6 | F_2-points + Bruhat-Tits 2-adic | DEEP (multi-week) | |

## 3. The integrated picture

The substrate's q=2 specialness is not just Mersenne integers — it's a STACK of characteristic-2 algebraic structures: Frobenius (layer 2), Steenrod (layer 3), modular reps (layer 5), 2-adic geometry (layer 6). Each layer has substrate-relevance hooks.

The substrate Hall algebra at q=2 is naturally INTEGER (over Z, evaluated at q=2 → Mersenne integers) — not modular per se. But the FRAMEWORK at q=2 is char-2-adjacent: Frobenius $x \to x^2$ is meaningful at q=2; the substrate's cohomology has Steenrod action; the restricted enveloping algebra is the modular analog. The substrate seems to live at a UNIQUE INTEGER LIFT of a char-2 structure.

## 4. Concrete program (v0.2 work, multi-week)

If we want to UNDERSTAND why q=2 specifically (beyond Mersennes), the multi-week investigation: 1. Layer 3 first (highest substrate-relevance, finite work): compute H^*(D_IV$^5$; F_2) and the Steenrod action. Identify substrate-natural Sq^i. 2. Layer 2 + 5: investigate the Hall coproduct's Frobenius behavior + compare restricted-enveloping simples to K-types. 3. Layer 6: 2-adic / F_2-points geometric structure (multi-week).

## 5. Honest scope + tier

RIGOROUS (existing math): Mersenne integers [n]_2 = M_n; Frobenius $x \to x^2$ in char 2; Steenrod algebra A_2; restricted enveloping algebra dim = p^(dim g); Lusztig quantum Frobenius (for root-of-unity case).

EXPLORATION (this v0.1): maps 6 layers; identifies substrate-relevance per layer; suggests concrete next steps. NOT a derivation; a research-direction map.

OPEN (multi-week): actual computation in each layer; substrate-specific derivations from each.

Cal #27 / honesty: this v0.1 is a TERRITORY MAP — it identifies WHERE substrate-relevance is, not WHAT the substrate-specific results would be. The Frobenius (layer 2) and Steenrod (layer 3) are most promising; the restricted enveloping (layer 5) connects to substrate K-types via dim-comparison; 2-adic (layer 6) is the deepest direction. Each layer has multi-week-substantive concrete next steps. v0.2+ would pick one and push.

Routed: → Elie: layer 3 (Steenrod on H^*(D_IV$^5$; F_2)) + layer 5 (restricted u(B_2)_{F_2} simple modules) are in your command lane (explicit computations). → Grace: catalog scan for restricted-enveloping modular reps + Bruhat-Tits / 2-adic substrate references. → Keeper: B1 v0.1 maps the territory; v0.2+ picks a layer (recommendation: layer 3 Steenrod, highest substrate-relevance with finite first computation). → me: continuing to Lyra Queue #3 = A3 L5 absolute scale framework.

— Lyra, B1 quantum groups at q=2 deeper v0.1. Six layers mapped: Mersenne (known), Char-2 Frobenius (HIGH relevance), Steenrod mod-2 cohomology (HIGH), quantum Frobenius vs root-of-unity (MODERATE; q=2 isn't a root of unity but has Mersenne periodicity), restricted enveloping u(B_2)_{F_2} dim 2^10=1024 (MED-HIGH; modular K-types?), F_2-points + 2-adic Bruhat-Tits (DEEP). Substrate's q=2 specialness is a STACK of char-2 structures, not just Mersennes. v0.2 picks one layer for explicit work (recommend layer 3 Steenrod, highest relevance + finite first computation).